

Calibo Task Name: Power BI Business Intelligence Dashboard & Data Modelling

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## EXECUTIVE BUSINESS INSIGHTS REPORT

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### 1. DATA OVERVIEW & METHODOLOGY

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#### Dataset Composition:

The analysed dataset contains the organization's comprehensive sales and operational history spanning 2021 to early 2023.

- \* Timeframe: 3 years of transactional data (2021, 2022, and Q1 2023).
- \* Products: 101 unique products featuring cost, original retail price, and tax data.
- \* Geography: 74 distinct locations, strictly concentrated across counties in California.
- \* Entities: 801 distinct Customers and a salesforce of 45 Salespeople.
- \* Financials: Monthly budgeting and revenue allocation targets per location.

#### Methodology & Actions Taken:

We executed an end-to-end Business Intelligence transformation workflow. The raw data was extracted, cleaned, and appended to merge three distinct years of sales into a single, unified 'Fact Sales' table. We designed and implemented a Star Schema Data Model, connecting the fact table to newly cleaned dimension tables (Products, Customers, Locations, Salespeople). A dedicated chronological Date table was engineered to support advanced DAX (Data Analysis Expressions) measures. We then calculated core metrics including Total Sales, Gross Profit, Margins, Year-over-Year (YoY) Growth, and Budget Variances. The final deliverable is a 7-page interactive Power BI Executive Dashboard featuring drill-down capabilities, cross-filtering, and dynamic slicers.

### 2. TOP 5 BUSINESS FINDINGS

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1. Strong Overall Profitability: The business generated \$25.66 Million in Total Sales across the full un-filtered dataset. Against total product costs, this yielded a Gross Profit of \$8.34 Million and a healthy overall Gross Margin of 32.52%.
2. Top Product Movers: Revenue is heavily driven by a few key performers. Product 63 and Product 28 are the highest revenue generators, bringing in over \$600,000 each in top-line sales.
3. The 2023 Data Limitation: While 2023 sales appear drastically lower than previous years (\$2.4M vs ~\$11.6M), dashboard time-intelligence reveals that 2023 only contains data for January, February, and March (Q1), meaning the business is actually maintaining its run-rate.
4. Budget Expectations Exceeded: Actual generated revenue (\$25.6M) vastly outperformed the recorded budget allocations (\$4.2M). The business achieved over 600% of its recorded budgetary targets.
5. Absolute Geographic Concentration: 100% of the recorded sales for this dataset were generated within the state of California across various counties (e.g., Los Angeles, Orange, Santa Clara).

### 3. TOP 5 BUSINESS PROBLEMS IDENTIFIED

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1. The "High-Revenue / Low-Margin" Trap: Product 84 is a top 5 revenue generator but has one of the worst profit margins in the entire catalogue (17%). It drives high volume but drains operational profitability.
2. Stagnant Year-Over-Year Growth: The business hit a plateau between 2021 and 2022. Sales slightly dropped from \$11.69M in 2021 to \$11.56M in 2022 (a ~1% decline), indicating a stall in market expansion.
3. Severe Budget Misalignment: The massive 600%+ variance between actual sales and the allocated budget indicates that the corporate forecasting models are severely flawed, making it difficult to allocate resources properly.
4. Critical Margin Drainers: Products like Product 4 and Product 95 are operating at critically low margins (under 16%), meaning selling these products barely covers wholesale and operational costs.
5. Over-Reliance on the California Market: Because the entire customer footprint resides in California, the revenue stream is highly vulnerable to regional economic downturns or state-specific supply chain disruptions.

### 4. TOP 5 BUSINESS RECOMMENDATIONS

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1. Optimize Pricing for Product 84: Immediately review the pricing strategy for Product 84. Management must either renegotiate supplier costs to lower the baseline expense or raise the retail price to bring its margin closer to the 32% company average.

2. Overhaul Financial Forecasting: The finance team must update the budgeting models for the upcoming fiscal year. Budgets should be set accurately against an \$11M+ annual run rate to ensure sales teams have realistic, motivating targets.
3. Incentivize High-Margin Sales: Adjust the Salesperson commission structure. Instead of rewarding raw sales volume, offer higher commission multipliers for moving products that yield profit margins greater than 40%.
4. Annualize 2023 Metrics for Accurate Reporting: When presenting to stakeholders, utilize the dashboard's Year-to-Date (YTD) DAX measures to compare Q1 2023 strictly against Q1 2022, preventing the false impression of a massive sales drop-off.
5. Geographic Market Expansion: Utilize the surplus profits generated in the California market to fund expansion into neighbouring states (such as Nevada, Oregon, or Arizona) to diversify geographic risk and reignite YoY growth.

